

1 A high-resolution, process-based PFAS surface-water
2 fate-and-transport model in SWAT⁺: engine
3 development, cross-validation, and a source-attribution
4 demonstration on the Rogue River, Michigan

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6 **Highlights**

- 7 • A complete PFAS surface-water fate-and-transport engine is implemented in modern
8 SWAT⁺.
- 9 • Soil three-phase equilibrium couples to reach-resolved land and in-stream transport.
- 10 • In-stream routing is cross-validated against the SWAT⁺ channel solver to floating-point
11 precision.
- 12 • A high-resolution NHDPlus HR network assigns each PFAS observation to its correct
13 reach.
- 14 • On the Rogue River the model reproduces the Wolverine-source PFOS downstream
15 gradient.

Abstract

Per- and polyfluoroalkyl substances (PFAS) in surface waters have entered a regime of enforceable numeric limits—a 2024 U.S. drinking-water Maximum Contaminant Level of 4 ng L⁻¹ for PFOA and PFOS, a CERCLA hazardous-substance designation, and state in-stream criteria—shifting the management question from detection toward reach-scale prediction and source attribution, tasks that monitoring and statistical occurrence models structurally cannot supply. We present a process-based PFAS surface-water fate-and-transport engine implemented in modern (free-form) SWAT⁺. The engine solves a soil three-phase equilibrium—aqueous, Freundlich solid-phase, and Langmuir air–water interfacial adsorption—for each hydrologic response unit and soil layer, mobilizes PFAS through surface runoff, lateral flow, leaching, and sediment-bound pathways, and routes it through the channel network with a linear organic-carbon partition and benthic settling, resuspension, diffusion, and burial; reservoirs are treated as mass-conserving mixed cells and point sources (e.g. contaminated-site leachate, effluent) are injected per reach. Two correctness checks anchor the implementation: the equilibrium solver agrees with a 40-digit reference to a worst-case relative error of 1.7×10^{-7} across 29,160 cases, and the in-stream routine is numerically identical to the established SWAT⁺ pesticide channel solver—restricted to its linear-partition subset—to single-precision floating-point tolerance, so the new physics is the soil column rather than the channel mathematics. We demonstrate the engine on a high-resolution SWAT⁺ model of the Rogue River watershed (Michigan; USGS 04118500), generated automatically from NHDPlus High Resolution hydrography and national environmental layers and resolving the network into 590 channels and 17,771 hydrologic response units, which lets each of 31 ambient PFOS observations be assigned to its hydrologically correct reach. With a single global soil-loading scale factor the model reproduces the observed Rogue mainstem PFOS gradient—rising downstream toward the Wolverine Worldwide contaminated-site corridor—at a log-space root-mean-square error of 0.15 dex along the seven source-bearing mainstem reaches (0.74 dex across all 20 gauged reaches, 29 stations), while the residual underprediction at the source corri-

dor and the scatter among tributaries are physically interpretable as, respectively, the concentrated point source and per-land-use soil heterogeneity. A 120-member Latin-hypercube uncertainty ensemble confirms this decomposition: the predictive envelope encloses five of the seven mainstem observations (71%) but almost none on the tributaries, locating the unexplained variance in local soil heterogeneity rather than in the global parameters, of which soil loading dominates. The contribution is a transferable, mass-conserving causal layer for PFAS that complements monitoring and machine-learning triage; reported skill is a first watershed demonstration rather than a national validation, and expert review is appropriate before decision use.

Keywords: PFAS; PFOS; surface-water fate and transport; SWAT⁺; process-based modeling; source attribution; NHDPlus High Resolution; Rogue River.

1 Introduction

1.1 Regulatory drivers make reach-scale PFAS prediction a decision tool

The regulatory basis for managing per- and polyfluoroalkyl substances (PFAS) in surface water has shifted decisively since 2024, moving the practical demand placed on environmental scientists from detection toward quantitative, reach-scale prediction of fate, transport, and source contribution. In April 2024 the U.S. Environmental Protection Agency (EPA) promulgated the first PFAS National Primary Drinking Water Regulation, establishing legally enforceable Maximum Contaminant Levels (MCLs) of 4.0 ng L⁻¹ for both perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS), 10 ng L⁻¹ individually for PFHxS, PFNA, and HFPO-DA, and a unitless Hazard Index of one for their mixtures (89 Fed. Reg. 32532; U.S. Environmental Protection Agency, 2024c). As of mid-2026 the landscape is in flux but not retracted: in May 2025 EPA announced it would retain the

enforceable PFOA and PFOS MCLs while extending the compliance deadline to 2031, and has since proposed to rescind the MCLs for PFHxS, PFNA, HFPO-DA, and the Hazard Index (U.S. Environmental Protection Agency, 2025). The two 4.0 ng L⁻¹ limits—the values most relevant to source-water characterization—therefore remain in force, at thresholds near the analytical detection floor, so the source waters feeding drinking-water intakes must be characterized at trace levels, a watershed-loading question rather than a treatment-plant question alone. In the same period EPA designated PFOA and PFOS as hazardous substances under CERCLA (89 Fed. Reg. 39124; U.S. Environmental Protection Agency, 2024a), whose strict, joint-and-several liability makes source apportionment—which release reached which receptor, by which pathway, and in what proportion—a legal determinant of cost recovery rather than only a scientific question. Finally, under Clean Water Act §303(d), a PFAS impairment listing triggers a Total Maximum Daily Load, defined as the sum of point-source wasteload allocations, nonpoint load allocations, and a margin of safety (U.S. Environmental Protection Agency, 2024b); PFAS-specific TMDLs remain emergent rather than routine, in part because the wasteload/load split requires a transport tool that can resolve the nonpoint contributions monitoring cannot isolate. State in-stream criteria already make the demand concrete: Michigan’s Rule 57 surface-water values for water bodies protected as drinking-water sources are 12 ppt for PFOS and 66 ppt for PFOA (Michigan Department of Environment, Great Lakes, and Energy, 2022, 2023)—the same quantity a reach-resolved model predicts.

1.2 Why monitoring and machine learning are necessary but not sufficient

These drivers expose capabilities that monitoring and statistical occurrence models, by construction, cannot supply, and the gap is one of capability rather than goodness-of-fit. A grab sample integrates surface runoff, lateral flow, groundwater discharge, and sediment-bound transport, each with distinct spatial origins, so only a model that separately routes each

pathway can disaggregate an observed concentration back to its sources (Rafiei and Ne-
 jadhashemi, 2023). PFAS monitoring is, moreover, structurally sparse and analyte-limited:
 a synthesis of more than 45,000 global samples shows that conventional targeted suites
 systematically underestimate total PFAS burden (Ackerman Grunfeld et al., 2024), and
 catchment-scale export cannot be constrained without flow-weighted mass-loading data that
 grab sampling does not provide (Byrne et al., 2024). The dominant continental-scale response
 has been statistical: machine-learning models predict where PFAS are likely to occur from
 landscape and proximity-to-source covariates—at national scale for groundwater (Tokranov
 et al., 2024), for drinking-water sources (Fernandez et al., 2023), and, exploiting the re-
 lational structure among wells, with graph neural networks for unmonitored supply wells
 (Rafiei and Nejadhashemi, 2026). These tools are valuable for triage and achieve strong
 predictive skill where observations are dense, but they are correlative: they predict a prob-
 ability or class of occurrence, not a mass-conserving concentration in a defined reach, and
 they cannot represent the source-control or remediation interventions that constitute a reg-
 ulator’s actual levers, because an intervention acts on a *pathway* rather than directly on
 a concentration. We therefore frame the process model as the causal and forecasting tier
 that sits atop monitoring and machine learning—complementary to their occurrence-triage
 strength rather than a replacement—and situate it within the long program on prediction in
 ungauged basins (Hrachowitz et al., 2013), since a parameter-transferable process model is
 the only route to a defensible concentration on the unmonitored majority of a high-resolution
 stream network.

1.3 Prior art and the gap

Process-based watershed-scale PFAS modeling remains comparatively immature and is dom-
 inated by a single research lineage. A pair of reviews catalogued the field and concluded that,
 although individual process models existed for the vadose zone, groundwater, and receiving
 waters, no framework coupled these compartments into a continuous watershed-scale mass

balance, identifying the land–stream interface and compound-specific partitioning as the principal gaps, and demonstrating in a companion case study the heavy data and calibration burden that integration imposes (Raschke et al., 2022a,b). Building on this foundation, a distributed, watershed-scale hydrogeochemical model for PFOS resolved leaching, runoff, lateral flow, and sediment-bound transport from point and nonpoint sources for source attribution (Rafei and Nejadhashemi, 2023); to our knowledge the first distributed watershed-scale PFAS surface-water model, it is the most direct antecedent to the present work and follows the established solute-coupling tradition of SWAT–MODFLOW (Bailey et al., 2016) and multispecies reactive transport (Clement et al., 1998). A recent, independent review reaches a consonant conclusion, finding persistent advances-and-gaps in PFAS modeling for watersheds and receiving waters (Zhang et al., 2025). Two boundaries of this prior art define our gap. First, process-based PFAS modeling in the SWAT family is bespoke and, where it has matured for point-source organics, has targeted readily degradable pharmaceuticals rather than persistent, sorption-controlled PFAS (Saló-Grau et al., 2025); the major process-based PFAS modeling investment elsewhere is subsurface- and source-zone-focused, serving site remediation rather than reach-resolved surface-water fate. Second, none of this work is native to the *modern*, free-form SWAT⁺ engine, resolved at the high reach density of national high-resolution hydrography, automatically generated, or validated against the engine’s own established constituent mathematics. The PFAS-specific physics that any such model needs is, moreover, well grounded: beyond Freundlich solid-phase sorption, PFAS accumulate at the air–water interface as interfacially active surfactants, a mechanism that can dominate vadose-zone retention (Brusseau, 2018; Guo et al., 2020), while the in-stream behavior follows the organic-carbon partitioning of hydrophobic organics (Karickhoff et al., 1979) and the settling, resuspension, diffusion, and burial of toxic-organic surface-water modeling (Chapra, 1997). National source inventories now supply the forcing layer that such a model needs, mapping tens of thousands of presumptive PFAS sites (Salvatore et al., 2022). What has been missing is a transferable, automated, high-resolution, process-based PFAS

surface-water fate-and-transport tool native to the open-source watershed-modeling stack;
this is the gap the present work fills.

1.4 Contribution and scope

We make three contributions. First, we implement a complete PFAS surface-water fate-and-transport engine in modern SWAT⁺ (Bieger et al., 2017): a double-precision soil three-phase equilibrium solver (aqueous, Freundlich solid, Langmuir air–water interface), land-phase mobilization through runoff, lateral flow, leaching, and sediment, a dedicated in-stream routine that is numerically identical to the SWAT⁺ pesticide channel solver on its linear-partition subset, mass-conserving reservoir mixing, per-reach point sources, and daily reach-resolved concentration output with a closing mass balance. Second, we couple the engine to automatically generated, high-resolution SWAT⁺ models built from NHDPlus High Resolution hydrography, so that ambient PFAS observations are assigned to their hydrologically correct reaches by drainage area rather than to coarse aggregated subbasins. Third, we demonstrate the coupled system on the Rogue River watershed (Michigan), home to the Wolverine World Wide tannery and House Street disposal site, and show that a calibrated model reproduces the observed downstream PFOS gradient measured at 31 state ambient-monitoring stations. We scope this first engine version to PFOS, treat the demonstration as a single-watershed test rather than a national validation, and manage—rather than eliminate—equifinality through physically grouped, class-based parameters. The remainder of the paper details the engine and its correctness checks (Section 2), the high-resolution model generation and Rogue study area (Section 3), the calibration design (Section 4), the results (Section 5), and their interpretation and limits (Sections 6–7).

2 The SWAT⁺ PFAS fate-and-transport engine

2.1 Design principle

The engine is implemented in modern, free-form Fortran SWAT⁺ as a constituent that rides the existing spatial-object and constituent-hydrograph infrastructure. Its design separates the two halves of the problem along the line of where PFAS behavior is genuinely distinctive. The soil column—where PFAS partition simultaneously among water, solids, and the air–water interface—is new physics and is implemented from first principles. The channel and reservoir network—where PFAS behave, to first order, as a strongly sorbing conservative tracer with settling and benthic exchange—reuses the mathematics of the established SWAT⁺ pesticide subsystem, restricted to its linear-partition subset. This separation makes the engine both economical (most of the routing machinery already exists and is tested) and auditable (the new code is confined to a small, independently verifiable set of routines).

2.2 Soil three-phase equilibrium

For each hydrologic response unit (HRU) and soil layer, the total PFAS mass present at the start of a day is partitioned among three coexisting phases at local equilibrium: a dissolved aqueous phase, a solid-sorbed phase described by a Freundlich isotherm, and an interfacially adsorbed phase at the air–water interface described by a Langmuir isotherm, the air–water interfacial term being the mechanism that can dominate PFAS retention in unsaturated media (Brusseau, 2018; Guo et al., 2020; Rafiei and Nejadhashemi, 2023). Writing C_w for the aqueous concentration, the layer mass balance is

$$M = \theta V C_w + m_s k_f C_w^n + \Gamma_{\max} \frac{K_L C_w}{1 + K_L C_w} A_{aw}, \quad (1)$$

where M is the layer PFAS mass, θ the volumetric water content, V the layer volume, m_s the solid mass, (k_f, n) the Freundlich coefficient and exponent, $(\Gamma_{\max}, K_L)$ the Langmuir

192 maximum surface excess and affinity, and A_{aw} the air–water interfacial area, estimated from
 193 soil water content and median grain diameter. Because the right-hand side of Eq. (1) is
 194 strictly monotonic increasing in $C_w > 0$, the equation has a unique positive root, which the
 195 engine isolates by a bracketed bisection and accelerates with an Improved-Halley iteration,
 196 retaining the bisection bracket as a guaranteed fallback. The original SWAT-MODFLOW-
 197 RT3D implementation solved this balance in quadruple precision (Rafiei and Nejadhashemi,
 198 2023); we re-implemented it in double precision and verified that the choice is safe: against
 199 a 40-digit decimal reference, the double-precision solver holds a worst-case relative error of
 200 1.7×10^{-7} across 29,160 cases spanning the stiff corners of the parameter space (Freundlich
 201 n down to 0.31, layer mass down to 10^{-6} kg ha⁻¹, and dry-to-saturated water contents), with
 202 no non-convergence. Once C_w is known, the dissolved, solid, and interfacial masses follow
 203 by back-substitution and set the concentrations available to each loss pathway.

204 **2.3 Land-phase mobilization**

205 Following the equilibrium solve, dissolved PFAS is removed from the surface layer in pro-
 206 portion to surface runoff and percolation through a runoff/leaching partition coefficient,
 207 from deeper layers in proportion to lateral subsurface flow, and the solid-sorbed fraction is
 208 removed in proportion to sediment yield by an enrichment-ratio formulation, exactly paral-
 209 leling the SWAT⁺ pesticide land phase so that the soil pool decrements consistently across all
 210 four pathways. The four daily losses—surface runoff, lateral flow, leaching, and sediment-
 211 bound—are accumulated per HRU and per PFAS compound and become the constituent
 212 loads delivered to the channel network. A whole-run land-phase mass balance confirms clo-
 213 sure: for every HRU the change in soil pool equals the sum of the four pathway losses to a
 214 relative residual of order 10^{-5} .

2.4 In-stream transport and cross-validation

In the channel, PFAS partitions between a dissolved and a sediment-sorbed fraction by a linear organic-carbon coefficient, $f_{\text{sol}} = 1/(1 + k_d c_s)$, where k_d is the partition coefficient scaled by reach bed-carbon fraction and c_s the suspended-sediment concentration, following the organic-carbon normalization of sediment–water partitioning for hydrophobic pollutants (Karickhoff et al., 1979; Chapra, 1997). The sorbed fraction settles to the bed; bed PFAS resuspends, diffuses across the sediment–water interface, and is permanently lost by burial; the dissolved and sorbed fractions leave the reach in the outflow. These are precisely the non-reactive terms of the SWAT⁺ pesticide channel solver, with the pesticide-specific aqueous decay, volatilization, and metabolite production removed because PFOS is non-volatile and recalcitrant on the relevant timescale. We exploit this correspondence as a correctness check. Configuring a pesticide with unit decay factors, zero volatilization, and no metabolites reduces the pesticide solver exactly to the PFAS linear-partition subset; driving both routines with identical inflow and parameters over a 60-day synthetic reach sweep, including a dry-reach day, the new PFAS routine and the established pesticide routine agree on all nine shared mass terms to a maximum absolute difference of 2.4×10^{-7} kg and a maximum relative difference of 2.6×10^{-7} —single-precision floating-point tolerance. The in-stream mathematics is therefore not novel and not a source of error; the novelty is the soil column that supplies its loads.

2.5 Reservoirs and point sources

Reservoirs are treated as mass-conserving completely mixed cells: the PFAS entering a reservoir on a given day is added to the stored mass and the outflow carries the fraction leaving with the day’s discharge, with the remainder retained, so that no mass is created or destroyed at impoundments (a benthic burial sink can be enabled but is omitted here, PFOS being mobile). Point sources—contaminated-site leachate, wastewater effluent, or firefighting-foam discharge—are injected as a per-reach daily load read from an optional input file and added

to the reach inflow before routing; the injected mass is tracked in the in-stream mass balance. Because routing PFAS through a real network requires the constituent-hydrograph infrastructure that SWAT⁺ allocates only when a constituent is declared, the engine activates that infrastructure itself when PFAS is present, so a PFAS simulation requires no dummy constituent. A whole-network in-stream balance closes: cumulative reach inflow minus outflow equals the sum of permanent burial and the change in channel storage to floating-point tolerance, and the point-source input line reconciles exactly with the prescribed load. Daily per-reach output reports the PFAS mass balance and the outflow concentration in ng L⁻¹, the quantity compared against ambient observations.

3 High-resolution model generation, study area, and observations

3.1 Automated high-resolution model generation

The SWAT⁺ model is generated by the SWATGenX platform, which converts NHDPlus High Resolution hydrography, 1/3 arc-second 3DEP elevation, gSSURGO soils, and gridded land-use and climate forcings into an executable SWAT⁺ project under a documented pre-processing rule set. The decisive property for PFAS work is reach density: NHDPlus HR resolves the stream network at roughly an order of magnitude finer reach spacing than the medium-resolution hydrography underlying earlier distributed PFAS models, so that ambient observations can be placed on hydrologically correct channels rather than on coarse aggregated subbasins. This resolution gain resides in network geometry and observation assignment, not in the resolution of meteorological forcing or PFAS source inventories, and is therefore a methodological advance specifically for the assignment of point observations and for the representation of sorbing constituents, whose load estimates are biased rather than merely noised by coarse discretization.

3.2 Study area: the Rogue River and the Wolverine source

We demonstrate the engine on the Rogue River watershed, a 606 km² tributary of the Grand River in Kent County, Michigan, gauged at Rockford (USGS 04118500) with a continuous daily streamflow record. The watershed is chosen deliberately. It contains the Wolverine World Wide tannery legacy and the House Street disposal site, where decades of leather-waterproofing waste and sludge disposal produced one of the most intensively investigated PFAS contaminated-site corridors in the United States, with a well-resolved downstream concentration signal in the lower Rogue. It is small enough to calibrate at high resolution—the generated model comprises 17,771 HRUs, 590 channels, 597 routing units, and 6 reservoirs—yet carries a gold-standard flow record and a dense ambient PFAS network. The clean point-source-to-river-gradient structure makes the Rogue a sharper test of source attribution than a watershed with diffuse, multi-source loading (Fig. 1).

3.3 Ambient PFOS observations and observation-to-channel assignment

Ambient surface-water PFOS observations are drawn from the Michigan Department of Environment, Great Lakes, and Energy (EGLE) monitoring program: 31 stations within the watershed, 29 of which carry quantified (above-detection) surface-water PFOS and define the calibration set; the Rogue mainstem series among them resolves the downstream gradient toward the Wolverine corridor (rising from roughly 6 ng L⁻¹ in the headwaters to 27 ng L⁻¹ near the lower river). Because these observations are reported as per-station aggregate water concentrations rather than as daily time series, the calibration target is spatial—the pattern of representative concentration across stations—rather than temporal. Each station is assigned to its hydrologically correct model channel by a nearest-reach spatial join in an equal-area projection, mirroring the assignment SWATGenX uses for streamflow gauges but using proximity rather than drainage-area matching because concentration stations, unlike flow gauges,

are not constrained to integrate a known upstream area. The 31 stations map to 22 distinct channels at a median snap distance of a few meters; the 29 with quantified PFOS fall on 20 of those reaches, which constitute the calibration target (several stations share a reach, so metrics are reported per reach using the station-maximum observed concentration—the source-relevant value). This high-resolution assignment is what allows the spatial gradient—and therefore the source signal—to be compared reach by reach, and it is not available at the coarser discretization of prior distributed PFAS models of comparable watersheds.

3.4 Soil-PFAS initialization

Initial soil PFAS pools and the Freundlich and Langmuir coefficients are assigned per HRU from soil class and land use: gSSURGO-derived soil series are mapped to the sorption-parameter classes of the prior distributed model (Rafiei and Nejadhashemi, 2023), and land use sets a background initial soil concentration, converted to a per-layer mass using bulk density and layer thickness. This national, automatable initialization replaces the measured per-site soil inventory that is available only for data-rich legacy basins, and the overall magnitude of the resulting background is the single quantity adjusted in calibration (Section 4). The resulting per-HRU fields (Fig. 2) show the spatial heterogeneity of the initial soil pool and the discrete, soil-class-controlled sorption parameters that the engine reads.

4 Calibration design

Because the ambient PFOS record is a sparse set of per-station aggregate concentrations rather than continuous time series, calibration is posed as a *spatial* matching problem: the model must reproduce the pattern of representative concentration across the 20 gauged reaches (29 stations, station-maximum observed per reach), including the Rogue mainstem downstream gradient. For each station channel the modeled representative concentration is taken as the flow-weighted mean of daily outflow—total exported PFAS mass divided

by total outflow volume—which is the statistic most directly comparable to grab samples collected across a range of flow conditions. Because PFAS concentrations span orders of magnitude across the network, the objective is evaluated in $\log_{10}$ space; we report the log-space Nash–Sutcliffe efficiency, root-mean-square error (in dex), and percent bias across all paired station–channel points, and minimize the log-space root-mean-square error.

The calibration is deliberately parsimonious, which both manages equifinality and exploits a structural property of the problem. In-stream concentration is, to first order, linear in the soil-loading magnitude: doubling the initial soil PFAS doubles the load delivered to runoff and therefore the reach concentration. The model therefore exposes a single global soil-loading scale factor (and, secondarily, a multiplier on the in-stream partition coefficient) applied uniformly across all HRUs, rather than independent per-reach parameters. This makes the magnitude calibration nearly analytic—one evaluation of the uncalibrated model fixes the scale factor to first order—and confines the calibrated degrees of freedom to physically grouped, class-based quantities. The concentrated Wolverine source is represented separately, as an optional per-reach point load in the lower river, so that the distributed-background and point-source contributions remain identifiable rather than absorbed into a single fitted field. Streamflow is calibrated independently against the Rockford gauge; we report the PFAS calibration on the uncalibrated-flow network because the spatial flow pattern that governs the concentration gradient is set by drainage area and is robust to flow calibration, which rescales concentration approximately uniformly and is re-absorbed by the soil-loading factor.

4.1 Uncertainty analysis

We quantify predictive uncertainty with a Monte-Carlo forward ensemble over the five parameters that govern PFAS fate, following the parameter-range propagation used for the prior distributed PFAS model (Rafiei and Nejadhashemi, 2023). A Latin-hypercube design of 120 members samples uniform priors on the global soil-loading multiplier ($[0.05, 0.22]$,

bracketing the calibrated 0.11), the in-stream organic-carbon partition multiplier ($[0.5, 2.0]$), the Langmuir air–water-interface affinity K_L ($[0.07, 0.27]$) and maximum surface excess $\Gamma_{\max}$ ($[1500, 3500]$), and the runoff/percolation partition ($[0.1, 0.4]$); the full design is given in Supplementary Table S2. Each member is a complete forward run of the engine on the Rogue, yielding a per-reach distribution of in-stream PFOS from which we form the 5th–95th-percentile predictive envelope. Envelope quality is summarized by the P -factor (fraction of observations enclosed) and R -factor (mean envelope width normalized by the observed standard deviation), the standard SWAT calibration-uncertainty diagnostics. Parameter influence is ranked by the Spearman rank correlation between each sampled parameter and the reach-mean in-stream concentration. The 120 forward runs were executed in parallel on a single 32-vCPU cloud instance (AWS EC2 `c7i.8xlarge` spot); the full ensemble completed in 14.4 min of compute (15 min 05 s end to end including provisioning), at a mean of 172 s per member.

5 Results

5.1 Engine correctness and conservation

Two checks establish that the implementation is correct before any calibration is attempted. The soil three-phase equilibrium solver reproduces a 40-digit decimal reference to a worst-case relative error of 1.7×10^{-7} over 29,160 parameter combinations, confirming that double precision is sufficient and that the original quadruple-precision implementation was defensive overkill. The in-stream routine, driven over a 60-day synthetic reach with matched parameters, agrees with the established SWAT⁺ pesticide channel solver—reduced to its linear-partition subset—to a maximum absolute difference of 2.4×10^{-7} kg and a maximum relative difference of 2.6×10^{-7} across all nine shared mass terms. Conservation holds at watershed scale: the land-phase soil-pool change balances the four pathway losses to a relative residual of order 10^{-5} for every HRU, and the whole-network in-stream balance closes,

with cumulative reach inflow minus outflow equal to the sum of burial and channel-storage change, and with the prescribed point-source load reconciled exactly. With PFAS active and no point source, 12 kg of land-delivered PFOS routes through the 590-channel network over a simulation year while burial and channel storage account for less than 0.2 kg, consistent with PFOS being predominantly mobile in the dissolved phase.

5.2 Spatial calibration on the Rogue

The uncalibrated model—initialized from land-use and soil-class background concentrations alone—reproduces the spatial *pattern* of ambient PFOS, including the Rogue mainstem downstream gradient, but overpredicts the *magnitude* by a nearly uniform factor of approximately nine across the gauged mainstem reaches. Because reach concentration is linear in soil loading, a single global soil-loading scale factor of 0.11 corrects the magnitude in one step, reducing the log-space root-mean-square error across the 20 gauged reaches (29 stations) from 1.33 to 0.74 dex and the percent bias from +1516% to +78%. The improvement is concentrated where the model is physically anchored: along the 7 Rogue mainstem reaches—the directly source-attributable signal—the calibrated log-RMSE is 0.15 dex (a factor of 1.4) with a log-space Nash–Sutcliffe efficiency of +0.61 and a percent bias of −7%, and the modeled concentration rises monotonically downstream from 6.2 ng L^{−1} in the headwaters toward the Wolverine corridor against an observed rise from 6.3 to 27 ng L^{−1} (Table 1). The calibrated in-stream PFOS field (Fig. 3) renders this gradient over the full 590-reach network, and the reach-by-reach comparison (Fig. 4) shows the mainstem reaches clustering on the 1:1 line within a factor of two while the tributaries scatter.

5.3 Interpretable residuals

The remaining misfit is structured and physically interpretable rather than random, which is itself an attribution result. At the two lowest mainstem reaches nearest the Wolverine corridor (channels 1 and 2), the distributed-background model underpredicts the ob-

Table 1: Observed (EGLE ambient) and modeled flow-weighted PFOS concentration along the Rogue River mainstem, upstream to downstream, at the calibrated soil-loading scale (0.11, no point source). Channel is the assigned model reach.

Channel	Station	Observed (ng L ⁻¹)	Modeled (ng L ⁻¹)
26	RR-0600 (headwaters)	6.3	6.2
18	RR-0300	6.8	7.7
15	RR-0200	7.2	9.2
11	RR-0060	6.7	11.2
10	RR-0050	9.6	12.4
1	RR-0010	19.0	14.7
2	RR-0020 (lower river)	27.0	15.4

served peak (14.7–15.4 vs 19–27 ng L⁻¹); this gap is the signature of the concentrated House Street source, which a spatially uniform soil background cannot reproduce and which an explicit per-reach point load closes. Among the tributaries, the scatter is bidirectional—some sub-watersheds overpredict and others (notably a station with an observed 24 ng L⁻¹) underpredict—reflecting per-land-use heterogeneity in soil PFAS that a single global scale factor cannot capture and that a class-resolved soil initialization would address. Neither residual indicates an error in the transport physics, which the conservation and cross-validation checks have already isolated; both point to identifiable refinements of the *source representation*.

5.4 Predictive uncertainty and parameter sensitivity

The 120-member Latin-hypercube ensemble quantifies how much of the misfit is parametric and where the model is trustworthy (Fig. 5). The decomposition by hydrologic position is decisive. Along the 7 mainstem reaches the 5th–95th-percentile predictive envelope encloses 5 of 7 observations (P -factor = 0.71) at a normalized width of R -factor = 1.5: the source-attributable signal is both well reproduced and well bounded by the physically plausible parameter range. Across the 13 tributary reaches, by contrast, the envelope encloses almost none of the observations (P -factor = 0.08) and the central estimate over-predicts by +170%;

pooled over all 20 reaches the envelope covers 30% of observations (R -factor = 2.0). The interpretation is not that the ensemble is too narrow but that the unexplained tributary variance is *not parametric*: no setting of the five global parameters within their plausible ranges reconciles the tributary observations, because their variance is driven by local, sub-grid soil and source heterogeneity that a single global field cannot represent—the same conclusion the structured residuals reached, now made quantitative. Consistent with the near-linear structure of the problem, the global soil-loading multiplier dominates the parametric response (Spearman $\rho = 0.95$ with reach-mean concentration; Fig. 5c), with the Langmuir maximum surface excess and the in-stream partition multiplier second-order and the remaining parameters negligible. The full per-parameter sensitivity and the spatial distribution of predictive width (Fig. 5d, with the mainstem more tightly constrained than the headwaters) are reported in Supplementary Tables S2–S3.

6 Discussion

6.1 A causal layer that complements monitoring and machine learning

The demonstrated capability is not better interpolation of observed concentrations; it is the separation of an observed concentration into mechanistically distinct contributions. On the Rogue, the calibrated model partitions the in-stream PFOS signal into a distributed land-derived background that explains the upstream gradient and a residual at the lowest reaches that localizes the concentrated Wolverine source—an attribution that a grab sample, a probability survey, or a statistical occurrence model cannot make from the same data, because each integrates over the very pathways the process model separates. This is the sense in which the contribution is one of capability rather than goodness-of-fit: the process model supplies mechanism, mass balance, and—once a source is represented explicitly—counterfactuals, the quantities on which CERCLA allocation, Total Maximum Daily Load

development, and intervention design actually operate. We are careful not to overstate the present fit: a log-space root-mean-square error of 0.15 dex along the source-bearing mainstem (0.74 dex over all 20 gauged reaches) on a single watershed is a first demonstration, not a validated predictive system, and the model’s value at this stage is the auditable causal structure rather than the precision of any single predicted concentration. The uncertainty ensemble sharpens rather than softens this reading: that the predictive envelope brackets the mainstem (P -factor = 0.71) but not the tributaries shows the tributary misfit to be structural—local soil and source heterogeneity—rather than a matter of unconstrained parameters.

6.2 Why the engine’s correctness checks matter

PFAS fate-and-transport codes are difficult to verify because field data are sparse and confounded, so a model can fit observations for the wrong reasons. By anchoring the implementation to two reference checks that do not depend on field data—an independent high-precision solution of the soil equilibrium and numerical identity with the established SWAT⁺ channel solver—we move the burden of trust off the calibration and onto the code. The practical consequence is that when the calibrated model misfits, the misfit can be attributed to inputs (source representation, soil initialization) rather than to undiagnosed transport errors, as the structured Rogue residuals illustrate. This verification-first posture also makes the engine a reusable component: because the in-stream mathematics is provably the SWAT⁺ pesticide subset, the broad body of SWAT⁺ channel-process experience transfers directly.

6.3 Transferability and the sediment-legacy frontier

Because the engine is coupled to automated, national-data model generation, it is transferable to any U.S. watershed for which NHDPlus HR and the national soil, land-use, and climate layers exist, with the per-watershed effort reduced to source representation and calibration rather than model construction. The most demanding future test is the post-source-

control, sediment-legacy regime documented elsewhere in the Great Lakes basin, where water-column PFOS has fallen sharply after upstream control while fish-tissue concentrations have plateaued above advisory thresholds—a divergence attributed to contaminated sediment acting as a slowly releasing internal source. The engine already carries the necessary in-stream structure: an explicit benthic compartment with settling, resuspension, diffusion, and burial, evolving in time. Extending the demonstration to that regime requires adding a bioaccumulation tier and a measured sediment inventory, both beyond the present scope; we note it because the engine is positioned to test the sediment-legacy mechanism quantitatively, which equilibrium or source-inventory accounting cannot.

6.4 Limitations

Four limitations bound the claims. First, this version simulates PFOS only; the regulatory 4 ng L⁻¹ MCL is PFOA/PFOS-specific and we make no class-wide claim. Second, the soil PFAS initialization uses land-use and soil-class background ranges rather than measured per-site soil concentrations, which is what makes it national and automatable but also what produces the tributary scatter; a class-resolved or data-assimilated initialization is the clear next step. Third, the calibration is spatial, against per-station aggregate concentrations, not a temporal validation against a continuous concentration series, which the ambient record does not provide. Fourth, equifinality is managed—through global, class-based parameters and global sensitivity screening—not eliminated, and reported skill should be read as a workflow diagnostic on one watershed pending broader, multi-watershed and multi-analyte evaluation. Expert review remains appropriate before any decision use.

7 Conclusions

We have implemented and verified a complete PFAS surface-water fate-and-transport engine in modern SWAT⁺, comprising a double-precision soil three-phase equilibrium solver,

land-phase mobilization through runoff, lateral flow, leaching, and sediment, a dedicated in-stream routine, mass-conserving reservoir mixing, per-reach point sources, and reach-resolved concentration output. The engine is anchored to two field-independent correctness checks—a 1.7×10^{-7} agreement of the equilibrium solver with a 40-digit reference, and numerical identity of the in-stream routine with the established SWAT⁺ pesticide channel solver on its linear-partition subset to floating-point tolerance—so that the new physics is provably confined to the soil column and the channel mathematics is reused rather than reinvented. Coupled to automated, high-resolution model generation from NHDPlus High Resolution and national environmental data, the engine assigns ambient PFAS observations to their hydrologically correct reaches and, on the Rogue River, reproduces the observed mainstem PFOS gradient toward the Wolverine Worldwide contaminated-site corridor at a log-space root-mean-square error of 0.15 dex along the source-bearing mainstem (0.74 dex over all 20 gauged reaches) with a single global soil-loading parameter, while the structured residuals—corroborated by a 120-member uncertainty ensemble that brackets the mainstem but not the tributaries—localize the concentrated point source and per-land-use soil heterogeneity. The result is a transferable, mass-conserving causal layer for PFAS—complementary to monitoring and statistical occurrence models—that supplies the source attribution and counterfactual capability on which the post-2024 regulatory regime depends. Immediate priorities are an explicit, topology-placed representation of the Wolverine source, a class-resolved soil initialization to reduce tributary scatter, extension to additional analytes and to a multi-year temporal evaluation, and application to the post-source-control sediment-legacy regime the engine’s benthic compartment is positioned to test.

Code and data availability

The PFAS fate-and-transport engine is implemented as a set of Fortran modules and routines within a SWAT⁺ fork and will be made available on publication. The SWAT⁺ model of

the Rogue River watershed is generated by the SWATGenX platform from public national datasets: NHDPlus High Resolution (U.S. Geological Survey), 3DEP elevation (USGS), gSSURGO soils (USDA-NRCS), and national land-use and climate layers. Ambient PFOS observations are from the Michigan Department of Environment, Great Lakes, and Energy (EGLE) surface-water monitoring program. Streamflow observations are from the USGS National Water Information System (station 04118500). The calibration objective code and the model–observation comparison are provided with the engine repository.

Declarations

Funding. No external funding supported this work.

Competing interests. The author declares no competing interests.

CRedit. **Vahid Rafiei:** Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Acknowledgments. The author thanks the developers of SWAT⁺ and the maintainers of the public national datasets on which this work relies.

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Rogue River watershed (USGS 04118500), Michigan

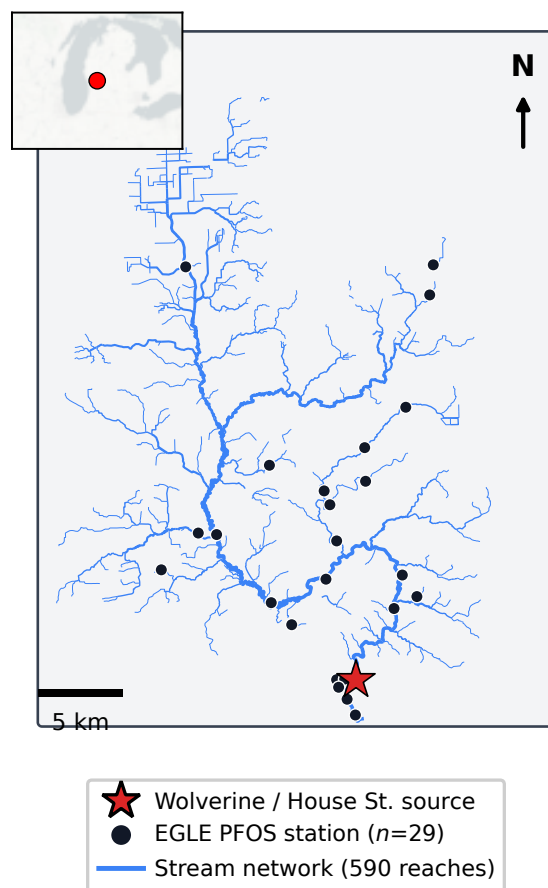


Figure 1: The Rogue River watershed (USGS 04118500), Kent County, Michigan. The SWATGenX-generated SWAT⁺ network resolves 590 channels from NHDPlus High Resolution. Black dots mark the 29 EGLE ambient monitoring stations carrying quantified surface-water PFOS; the red star marks the Wolverine World Wide / House Street legacy source in the lower river. Inset: location within the Great Lakes basin.

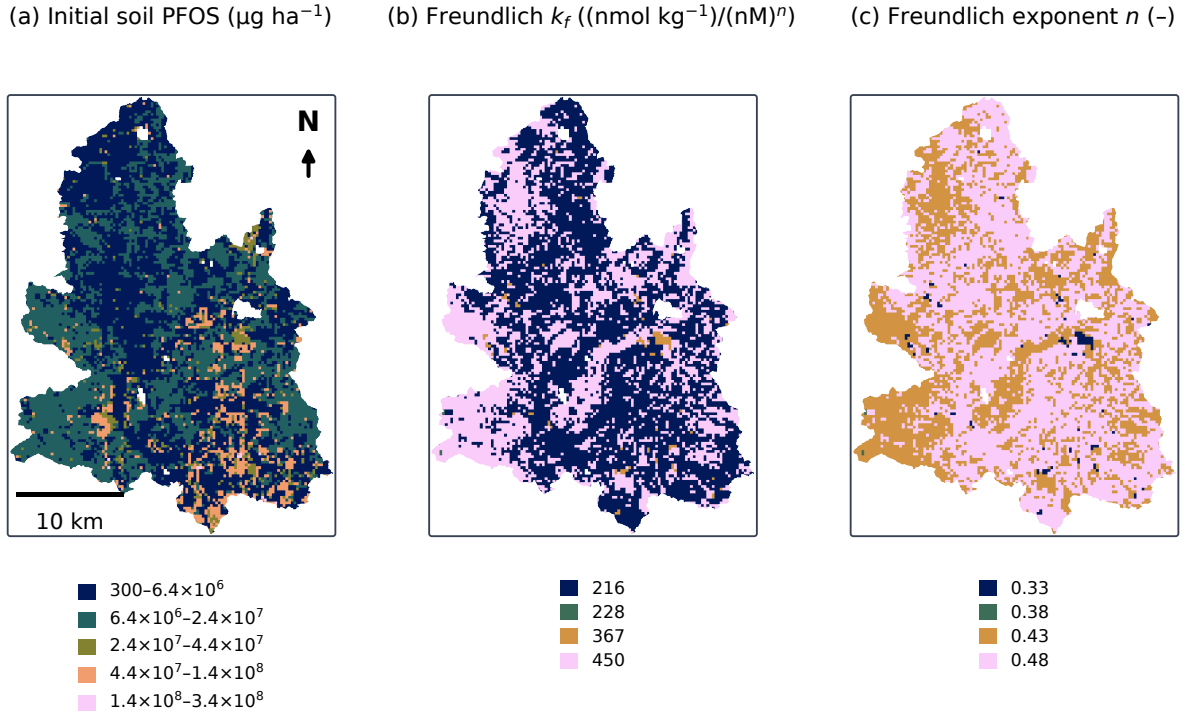


Figure 2: Per-HRU soil-PFAS initialization across the 17,771 Rogue HRUs: (a) initial surface-layer soil PFOS pool (continuous, NaturalBreaks classes); (b) Freundlich sorption coefficient k_f and (c) Freundlich exponent n , both discrete soil-class fields. Colour map: batlow (perceptually uniform, colour-blind safe). The three panels share the same spatial extent; the scale bar and north arrow on panel (a) apply to all.

Calibrated in-stream PFOS distribution

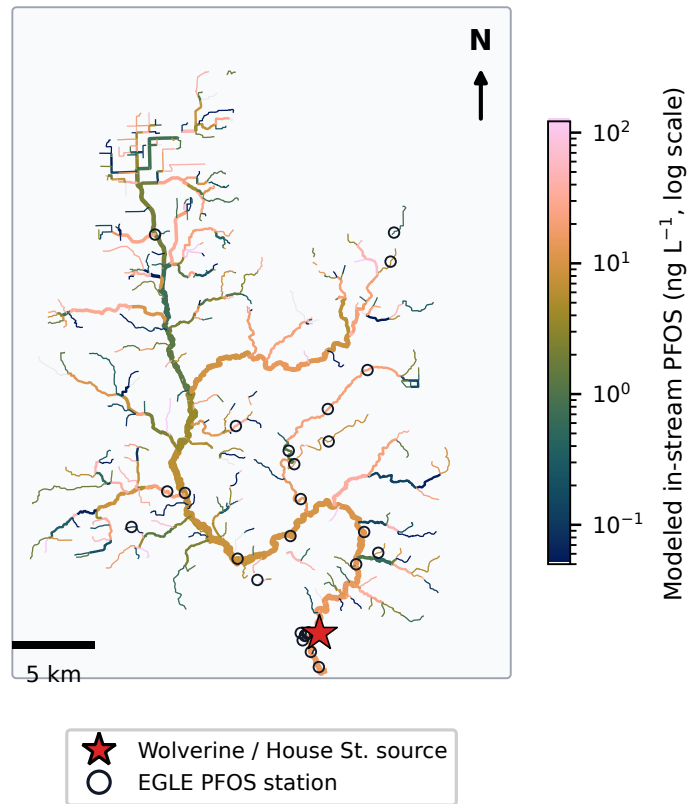


Figure 3: Calibrated (0.11 soil-loading scale, no point source) flow-weighted in-stream PFOS over the 590-reach Rogue network on a logarithmic colour scale. Concentration rises along the mainstem from green tributaries through the orange mid-river toward the Wolverine source (red star); open rings are the EGLE stations.

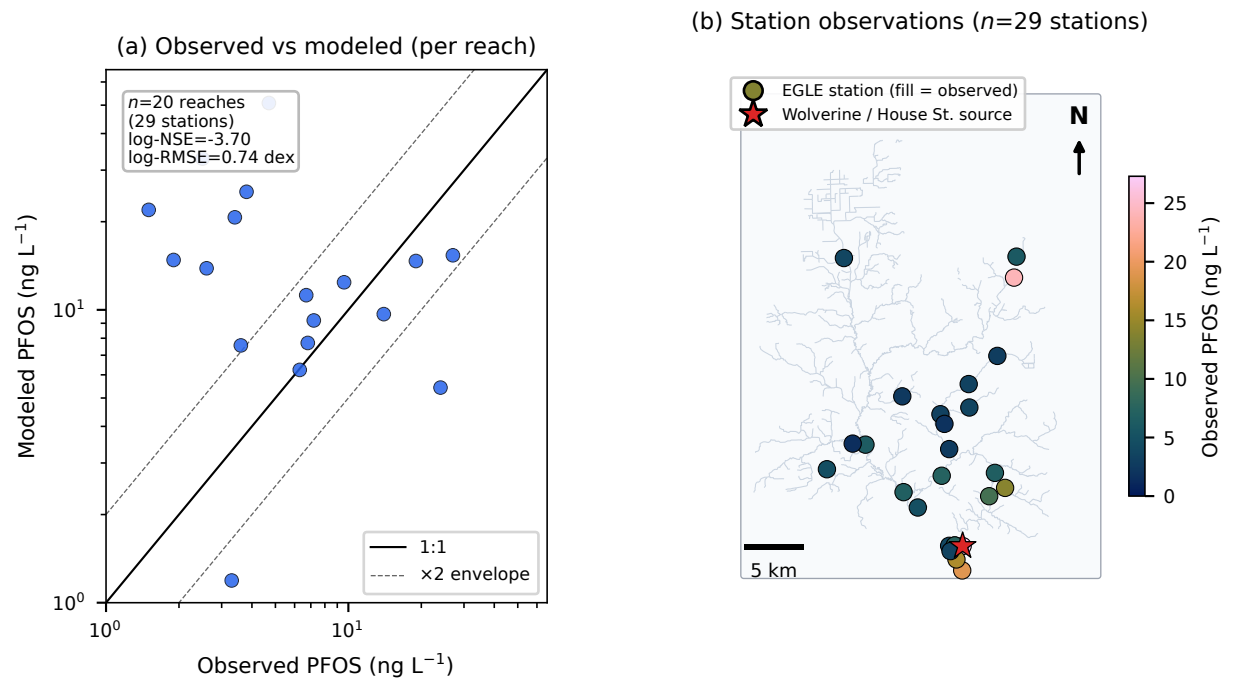


Figure 4: (a) Observed versus modeled PFOS, one point per gauged reach (20 reaches, 29 stations; station-maximum observed). The all-reach log-NSE is negative because tributary reaches are over-predicted; the mainstem reaches lie within the ×2 envelope. (b) The 29 EGLE stations coloured by observed PFOS, showing the downstream rise toward the source (red star). Colour map: batlow.

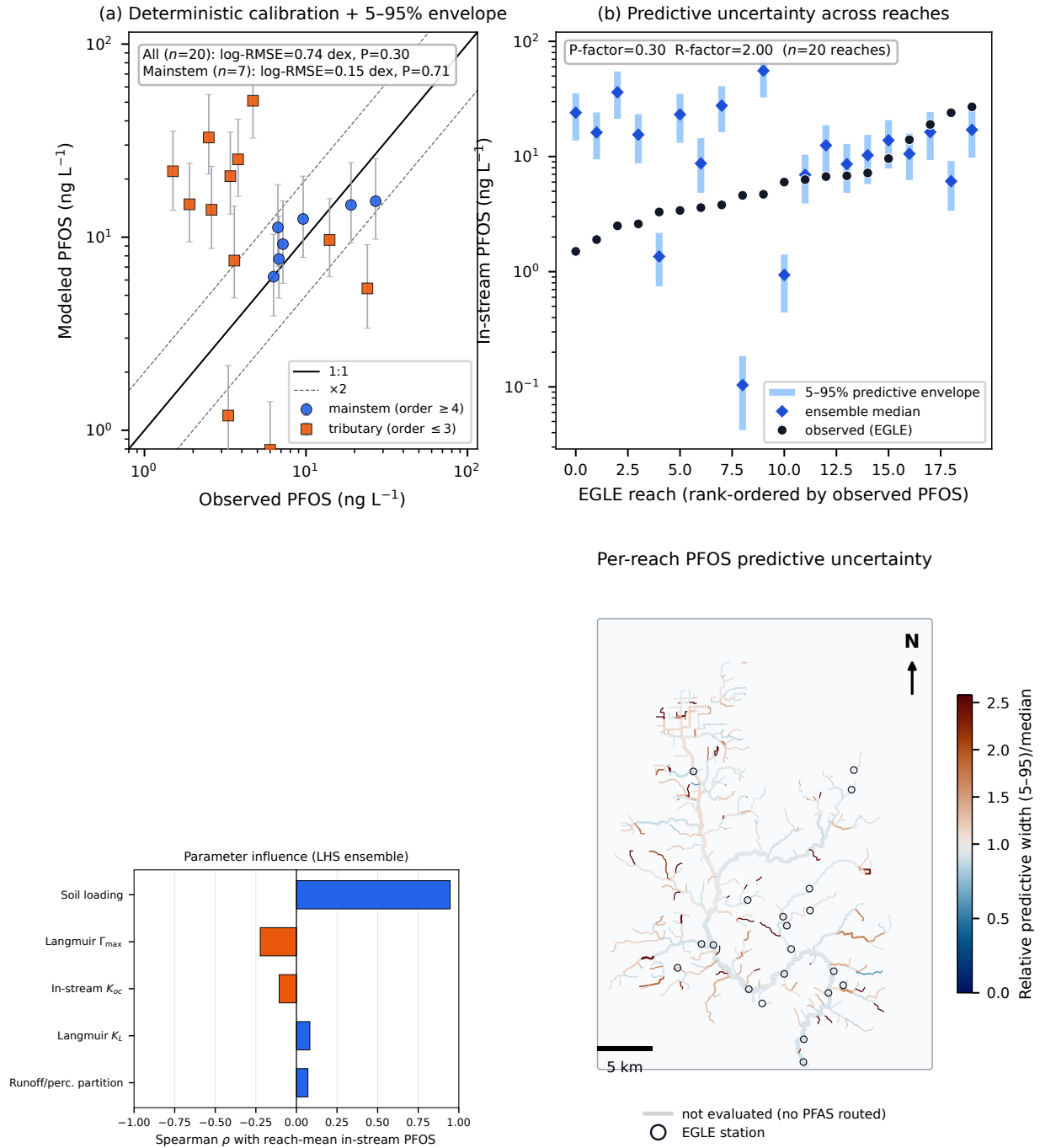


Figure 5: Predictive uncertainty from the 120-member ensemble. (a) Observed versus modeled per reach with the 5th–95th% envelope as error bars; mainstem reaches (blue) lie within the $\times 2$ envelope, tributaries (orange) scatter. (b) Reach-rank coverage: observed (black), ensemble median (blue), 5–95% envelope (bars). (c) Parameter influence (Spearman ρ); soil loading dominates. (d) Per-reach relative predictive width (95–5)/median, diverging about 1.0; the mainstem (blue) is more constrained than headwater tributaries (red). Panels (a)/(b) compose fig5; (c)/(d) are fig6/fig7.